From Maxwell's Equations to Lithographic Imaging

Conventions Through The Document. Time dependence is $e^{-i\omega t}$, λ_0 is the vacuum wavelength, and $\lambda_m = \lambda_0/n$ is the wavelength in a lossless medium. Transverse coordinates are $\boldsymbol{x} = (x, y)$; $\boldsymbol{f} = (f_x, f_y)$ is spatial frequency in *cycles per unit length*. Except where a refractive index is explicitly retained, the diffraction and imaging calculations below use vacuum or air, with $n = 1$, $\lambda = \lambda_0$, and $k = 2\pi/\lambda$.

1 From Maxwell's equations to the intensity we measure

1.1 The wave equation and the meaning of a complex field

In a linear, isotropic medium, $\boldsymbol{D} = \epsilon \boldsymbol{E}$ and $\boldsymbol{B} = \mu \boldsymbol{H}$. Maxwell's equations for the *real* fields are

$$\begin{aligned} \nabla \cdot (\epsilon \boldsymbol{E}) &= \rho_{\text{free}}, & \nabla \cdot (\mu \boldsymbol{H}) &= 0, \\ \nabla \times \boldsymbol{E} &= -\mu \partial_t \boldsymbol{H}, & \nabla \times \boldsymbol{H} &= \boldsymbol{J}_{\text{free}} + \epsilon \partial_t \boldsymbol{E}. \end{aligned} \tag{1.1}$$

The physical source involves charges and currents. In a homogeneous, source-free propagation region, set $\rho_{\text{free}} = 0$, $\boldsymbol{J}_{\text{free}} = 0$, and take ϵ, μ constant. Taking a curl,

$$\begin{aligned} \nabla \times (\nabla \times \boldsymbol{E}) &= -\mu \partial_t (\nabla \times \boldsymbol{H}) = -\mu \epsilon \partial_t^2 \boldsymbol{E}, \\ \nabla(\nabla \cdot \boldsymbol{E}) - \nabla^2 \boldsymbol{E} = -\mu \epsilon \partial_t^2 \boldsymbol{E} \quad &\Longrightarrow \quad \boxed{\nabla^2 \boldsymbol{E} - \mu \epsilon \partial_t^2 \boldsymbol{E} = 0.} \end{aligned} \tag{1.2}$$

For one optical frequency, introduce a spatial phasor $\widetilde{\boldsymbol{E}}$:

$$\boldsymbol{E}(\boldsymbol{r}, t) = \text{Re}\{\widetilde{\boldsymbol{E}}(\boldsymbol{r}) e^{-i\omega t}\}, \qquad \boxed{(\nabla^2 + k_m^2)\widetilde{\boldsymbol{E}} = 0, \quad k_m = \omega\sqrt{\mu\epsilon} = \frac{2\pi n}{\lambda_0}.} \tag{1.3}$$

The second time derivative supplies $-\omega^2$. Equation (1.3) is the vector Helmholtz equation. Its components satisfy scalar Helmholtz equations in this homogeneous region, but also obey the Maxwell constraint $\nabla \cdot \widetilde{\boldsymbol{E}} = 0$. For a common, approximately fixed polarization we use one scalar phasor $\mathcal{E}$ through $\widetilde{\boldsymbol{E}} = \widehat{\boldsymbol{e}}\,\mathcal{E}$. This scalar imaging approximation needs revision when polarization and longitudinal-field effects matter.

1.2 Why is it that $I \propto |\mathcal{E}|^2$

Intensity is averaged energy flux, so begin with $\boldsymbol{S} = \boldsymbol{E} \times \boldsymbol{H}$. For a travelling plane wave in a lossless medium, Faraday's law relates its phasors:

$$\boldsymbol{k} \times \widetilde{\boldsymbol{E}} = \omega\mu \widetilde{\boldsymbol{H}}, \qquad \widetilde{\boldsymbol{H}} = \frac{1}{\eta} \widehat{\boldsymbol{s}} \times \widetilde{\boldsymbol{E}}, \qquad \eta = \sqrt{\mu/\epsilon}. \tag{1.4}$$

Here $\widehat{\boldsymbol{s}}$ is the propagation direction and $\widehat{\boldsymbol{s}} \cdot \widetilde{\boldsymbol{E}} = 0$. For linear polarization, $\mathcal{E} = A e^{i\phi}$ gives the particularly transparent check

$$E = A\cos(\phi - \omega t), \qquad S_{\parallel} = \frac{A^2}{\eta}\cos^2(\phi - \omega t), \qquad \boxed{I = \left\langle S_{\parallel} \right\rangle_{\text{optical}} = \frac{A^2}{2\eta} = \frac{|\mathcal{E}|^2}{2\eta}.} \tag{1.5}$$

Thus the factor $1/2$ comes from the optical-cycle average, and $1/\eta$ from the magnetic-to-electric-field relation. Equivalently, using $E = (\mathcal{E} e^{-i\omega t} + \mathcal{E}^* e^{i\omega t})/2$,

$$\left\langle E^2 \right\rangle_{\text{optical}} = \frac{1}{4}\left\langle \mathcal{E}^2 e^{-2i\omega t} + 2|\mathcal{E}|^2 + (\mathcal{E}^*)^2 e^{2i\omega t} \right\rangle = \frac{1}{2}|\mathcal{E}|^2.$$

These are *peak* phasor amplitudes; an RMS amplitude already includes the factor $1/\sqrt{2}$. In a nonmagnetic dielectric, $1/\eta = n\epsilon_0 c$, so $I = n\epsilon_0 c|\mathcal{E}|^2/2$. The general monochromatic flux formula is $\langle \boldsymbol{S} \rangle_{\text{optical}} = \frac{1}{2}\,\mathrm{Re}(\widetilde{\boldsymbol{E}} \times \widetilde{\boldsymbol{H}}^*)$. It remains the appropriate starting point for vector, absorbing, or standing-wave fields where the simple travelling-wave reduction is inadequate.

1.3 Exactly how E and U are linked

We use an intensity-normalized scalar amplitude throughout the rest of these notes:

$$\boxed{U = \frac{\mathcal{E}}{\sqrt{2\eta}}, \qquad E(\boldsymbol{r}, t) = \mathrm{Re}\{\sqrt{2\eta}\, U(\boldsymbol{r}) \mathrm{e}^{-\mathrm{i}\omega t}\}, \qquad I = |U|^2.} \tag{1.6}$$

Here E is the real field component along the chosen linear polarization. $\mathcal{E}$ is measured in V/m; U is measured in $\sqrt{\mathrm{W}}/\mathrm{m}$. Multiplying by a constant does not change the homogeneous Helmholtz equation, so U satisfies it too. An alternative convention calls the unscaled phasor itself U, in which case $I = |U|^2/(2\eta)$. The convention must be stated once and kept consistent.

For a fluctuating narrowband source, write a slowly varying amplitude $U(\boldsymbol{r}, \tau)$ and then measure $I = \langle |U|^2 \rangle_{\text{exposure}}$. Optical-cycle averaging and averaging relative-phase fluctuations are separate steps; the latter determines coherence. A source can therefore be described downstream by its field or its field correlations on a chosen plane, without solving its microscopic emission process each time. See Born–Wolf and Saleh–Teich for the electromagnetic and scalar-wave foundations [1, 2].

1.4 Helmholtz equation

We can now make the transition from the real electric field to the spatial equation for U explicit. Retain a homogeneous, source-free, linear, isotropic, lossless medium, with constant ϵ and μ. The wave equation applies to each Cartesian electric-field component; representing the whole field by one scalar additionally uses the polarization approximation described above.

1. Start from the time-dependent Maxwell equation. Taking the curl of Faraday's law and using the source-free Ampere law gives $\nabla \times (\nabla \times \boldsymbol{E}) = -\mu\epsilon\partial_t^2 \boldsymbol{E}$. Since constant ϵ and $\nabla \cdot (\epsilon \boldsymbol{E}) = 0$ imply $\nabla \cdot \boldsymbol{E} = 0$, the curl–curl identity gives, for a real component E,

$$\boxed{\nabla^2 E - \frac{1}{v^2}\frac{\partial^2 E}{\partial t^2} = 0,} \qquad v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{n}. \tag{1.7}$$

2. Separate the single-frequency time dependence. For a monochromatic field, define the complex representation

$$E_{\mathrm{c}}(\boldsymbol{r}, t) = \mathcal{E}(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\omega t} = \sqrt{2\eta}\, U(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\omega t}, \qquad E = \mathrm{Re}\, E_{\mathrm{c}}. \tag{1.8}$$

The subscript c marks the complex representation. The phasor $U(\boldsymbol{r})$ has no time dependence for this one frequency, but its amplitude and phase may both vary with position.

3. Evaluate the derivatives separately. Because η is spatially constant in this medium,

$$\begin{aligned} \nabla^2 E_{\mathrm{c}} &= \sqrt{2\eta}\, \mathrm{e}^{-\mathrm{i}\omega t} \nabla^2 U, \\ \frac{\partial^2 E_{\mathrm{c}}}{\partial t^2} &= \sqrt{2\eta}\, U(-\mathrm{i}\omega)^2 \mathrm{e}^{-\mathrm{i}\omega t} = -\omega^2 \sqrt{2\eta}\, U \mathrm{e}^{-\mathrm{i}\omega t}. \end{aligned} \tag{1.9}$$

The Laplacian differentiates the spatial phasor; the time derivatives act only on the exponential term.

4. Substitute and cancel the common factor. Insert Eq. (1.9) into Eq. (1.7), using its complex representation:

$$0 = \sqrt{2\eta}\, \mathrm{e}^{-\mathrm{i}\omega t} \left[\nabla^2 U + \frac{\omega^2}{v^2} U \right].$$

The prefactor is nonzero, so the bracket must vanish. Define the wave number in the medium by

$$k_m = \frac{\omega}{v} = \frac{n\omega}{c} = nk_0 = \frac{2\pi}{\lambda_m}, \qquad k_0 = \frac{\omega}{c} = \frac{2\pi}{\lambda_0}. \tag{1.10}$$

We obtain the scalar Helmholtz equation,

$$\boxed{\nabla^2 U + k_m^2 U = 0, \qquad \text{or} \qquad \nabla^2 U + n^2 k_0^2 U = 0.} \tag{1.11}$$

For example, in Cartesian coordinates this means $\partial_x^2 U + \partial_y^2 U + \partial_z^2 U + k_m^2 U = 0$. The unscaled phasor $\mathcal{E} = \sqrt{2\eta}\, U$ obeys the same equation because the normalization is constant.

5. Check the result with a plane wave. Let $U = U_0 \mathrm{e}^{\mathrm{i}(k_x x + k_y y + k_z z)}$. Each second spatial derivative supplies a minus sign, giving

$$\nabla^2 U = -(k_x^2 + k_y^2 + k_z^2) U \quad \Longrightarrow \quad k_x^2 + k_y^2 + k_z^2 = k_m^2. \tag{1.12}$$

For a propagating plane wave with a real wavevector, $|\boldsymbol{k}| = k_m = 2\pi/\lambda_m$, as expected.

2 After the source: pupils, spatial frequencies, and propagation

2.1 What each part of the optical system does



The aperture stop limits the accepted ray cone. Its images through the preceding and following optics are the *entrance* and *exit* pupils. For an image-side cone of half-angle $\alpha_{\max}$,

$$\mathrm{NA} = n \sin \alpha_{\max}, \qquad f_c = \frac{\mathrm{NA}}{\lambda_0}. \tag{2.1}$$

NA is dimensionless; f_c is a field spatial-frequency cutoff. The illumination optics determine which directions reach the mask. The projection pupil determines which outgoing components the projector admits. A source distribution and a projection pupil are different objects.

For a thin scalar mask with complex amplitude transmission or reflection $t(\boldsymbol{x})$, the field just after the mask is

$$U_o(\boldsymbol{x}) = t(\boldsymbol{x}) U_{\mathrm{inc}}(\boldsymbol{x}). \tag{2.2}$$

Later, the imaging optics apply their own amplitude and phase response. Mask reflectance belongs in t; projection-mirror losses belong in the system response. They do not count the same reflection twice. Setting $U_{\mathrm{inc}} = 1$ assumes a spatially constant coherent field and chooses its amplitude as the unit. *Normalization, uniformity, and coherence are independent concepts.*

2.2 The pupils are the apparent stop

Imagine looking into the two-lens system in Fig. 1 from each end. From the **left**, lens L_1 makes the stop appear larger and at a different position: this apparent opening is the **entrance pupil (EP)**.

From the **right**, lens L_2 gives another apparent size and position: the **exit pupil (XP)**. Both are images of the same physical stop [1, 2].

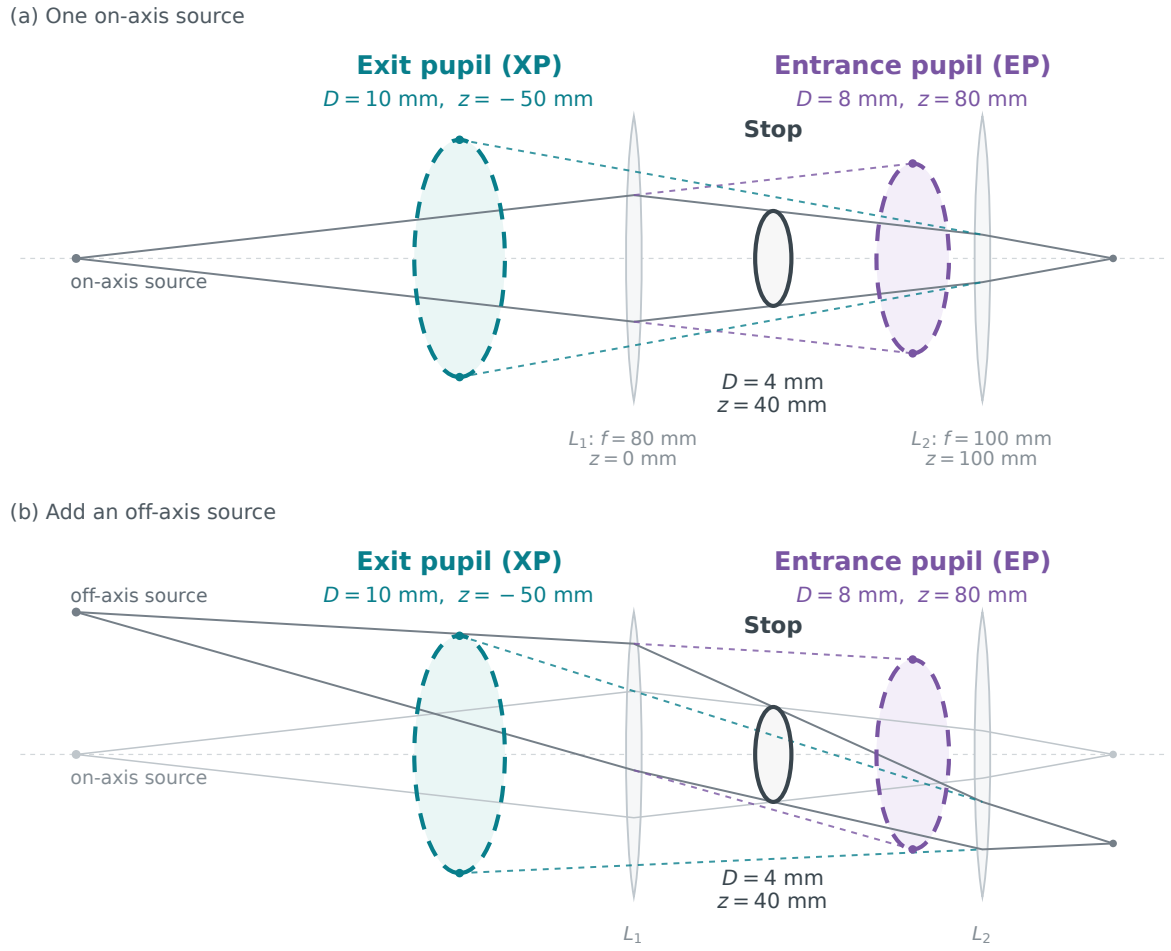


Figure 1: Pupil position and diameter D for (a) one on-axis source and (b) an added off-axis source. Thin solid rays pass through the stop. Purple dashed extensions of the incoming rays meet the EP edges; teal dashed extensions of the outgoing rays meet the XP edges. The coloured openings are virtual images of the physical stop. Both sources share the same pupils. Diameters share one vertical scale; axial distances use a different scale.

The names *entrance* and *exit* refer to the viewing side.

The entrance pupil describes the opening seen from the mask. The exit pupil's diameter and distance from the wafer determine the image-side cone angle, and hence the Numerical Aperture and cutoff spatial frequency.

2.3 A spatial frequency is variation across a fixed plane

Use the transverse Fourier pair

$$\widetilde{U}(\boldsymbol{f}, z) = \int U(\boldsymbol{x}, z) \mathrm{e}^{-\mathrm{i}2\pi \boldsymbol{f}\cdot\boldsymbol{x}} \mathrm{d}^2 x, \qquad U(\boldsymbol{x}, z) = \int \widetilde{U}(\boldsymbol{f}, z) \mathrm{e}^{\mathrm{i}2\pi \boldsymbol{f}\cdot\boldsymbol{x}} \mathrm{d}^2 f. \tag{2.3}$$

For a plane wave tilted towards x by θ in a lossless medium,

$$U(x, z) = U_0 \exp\left[\mathrm{i}\frac{2\pi n}{\lambda_0}(x \sin\theta + z\cos\theta)\right] \quad \Longrightarrow \quad f_x = \frac{n\sin\theta}{\lambda_0}.$$

Larger tilt means faster phase variation across the *fixed* $z =$ constant plane.

A fine mask pattern has large spatial frequencies in their transmittance/reflectance. Under normal, uniform illumination these also appear in U_o. Oblique illumination $\mathrm{e}^{\mathrm{i}2\pi \boldsymbol{s}\cdot\boldsymbol{x}}$ shifts the field spectrum to $\widetilde{t}(\boldsymbol{f}-\boldsymbol{s})$. The intensity $|U|^2$ has its own frequencies, which are differences between field frequencies.

2.4 Exact propagation and evanescent components

Fourier-transform the transverse derivatives in Helmholtz equation:

$$\partial_z^2 \widetilde{U} + \left[k_m^2 - (2\pi)^2 |\boldsymbol{f}|^2\right] \widetilde{U} = 0. \tag{2.4}$$

For outgoing propagation into $z > 0$, choose the forward or decaying branch:

$$\begin{aligned} &\widetilde{U}(\boldsymbol{f}, z) = H_z(\boldsymbol{f}) \widetilde{U}(\boldsymbol{f}, 0), \\ &H_z(\boldsymbol{f}) = \mathrm{e}^{\mathrm{i}k_z z}, \qquad k_z = \sqrt{k_m^2 - (2\pi)^2 |\boldsymbol{f}|^2}, \quad \operatorname{Re} k_z \geq 0, \quad \operatorname{Im} k_z \geq 0. \end{aligned} \tag{2.5}$$

Substituting $k_m = 2\pi n/\lambda_0$ gives the **exact propagation transfer function** explicitly:

$$\boxed{H_z(f_x, f_y) = \exp\left[\mathrm{i}2\pi z \sqrt{\left(\frac{n}{\lambda_0}\right)^2 - f_x^2 - f_y^2}\right].} \tag{2.6}$$

In vacuum, $n = 1$, $\lambda = \lambda_0$, and $k = 2\pi/\lambda$, so this is

$$H_z(\boldsymbol{f}) = \exp\left[\mathrm{i}kz\sqrt{1 - \lambda^2 |\boldsymbol{f}|^2}\right].$$

The full square root is retained. Multiplying the input spectrum by H_z and taking the inverse transverse Fourier transform gives

$$U(\boldsymbol{x}, z) = \mathcal{F}^{-1}\left[H_z(\boldsymbol{f}) \widetilde{U}(\boldsymbol{f}, 0)\right](\boldsymbol{x}). \tag{2.7}$$

For $|\boldsymbol{f}| < n/\lambda_0$, k_z is real: the component propagates, and H_z changes its phase. For $|\boldsymbol{f}| > n/\lambda_0$,

$$k_z = \mathrm{i}\kappa, \qquad \kappa = 2\pi\sqrt{|\boldsymbol{f}|^2 - (n/\lambda_0)^2}, \qquad H_z = \mathrm{e}^{-\kappa z}. \tag{2.8}$$

This is an evanescent component, with amplitude decay length $1/\kappa$. It still satisfies $\boldsymbol{k} \cdot \boldsymbol{k} = k_m^2$; the wavevector is complex, so the real-vector bound $|\boldsymbol{k}_\perp| \leq k_m$ no longer applies. The equality $|\boldsymbol{f}| = n/\lambda_0$ is the grazing boundary, $k_z = 0$.

Transverse frequency	Meaning for a clear circular pupil
$\lvert\boldsymbol{f}\rvert \leq f_c$	Within the pupil's angular acceptance.
$f_c < \lvert\boldsymbol{f}\rvert < n/\lambda_0$	Propagating, but rejected by this pupil.
$\lvert\boldsymbol{f}\rvert > n/\lambda_0$	Evanescent in this medium.

A sharp edge requires a wide range of field frequencies: some propagate and are rejected by the pupil, while sufficiently high ones are confined near the edge. Both mechanisms limit transferred detail, for different reasons.

2.5 Fresnel propagation is an approximation to this result

In vacuum, expand the square root for $\lambda|\boldsymbol{f}| \ll 1$:

$$k_z = k\sqrt{1-\lambda^2|\boldsymbol{f}|^2} \simeq k - \pi\lambda|\boldsymbol{f}|^2,$$
$$H_z(\boldsymbol{f}) \simeq \mathrm{e}^{\mathrm{i}kz}\mathrm{e}^{-\mathrm{i}\pi\lambda z|\boldsymbol{f}|^2}. \tag{2.9}$$

Inverse transformation gives the convolution with a quadratic-phase kernel:

$$\boxed{U(\boldsymbol{x},z) = \frac{\mathrm{e}^{\mathrm{i}kz}}{\mathrm{i}\lambda z}\int U(\boldsymbol{x}',0)\exp\left[\frac{\mathrm{i}\pi|\boldsymbol{x}-\boldsymbol{x}'|^2}{\lambda z}\right]\mathrm{d}^2x'.} \tag{2.10}$$

The plane called $z = 0$ is whichever input plane has a known field. Free propagation applies between optical elements. Reflection, refraction, masks, and coatings require their own boundary conditions or transfer factors. In a folded mirror train, a local axis may be chosen for each propagation segment.

The scalar angular spectrum is exact for its homogeneous, source-free model, not for a complete scanner. Fresnel propagation is useful for suitable paraxial segments; *high-angle fields and the material interfaces of an EUV mask can require vector Maxwell calculations.*

3 How focusing and aberrations enter the point-spread function

3.1 The three phase terms that cancel at focus

Consider a thin lens in air with focal length F, object distance s_o, image distance s_i, and pupil coordinate $\boldsymbol{\xi}$. For an on-axis object and on-axis observation point, the pupil-dependent phases are

$$\underbrace{\frac{k|\boldsymbol{\xi}|^2}{2s_o}}_{\text{object to lens}} \quad \underbrace{-\frac{k|\boldsymbol{\xi}|^2}{2F}}_{\text{lens}} \quad \underbrace{+\frac{k|\boldsymbol{\xi}|^2}{2s_i}}_{\text{lens to image}} = \frac{k|\boldsymbol{\xi}|^2}{2}\left(\frac{1}{s_o}+\frac{1}{s_i}-\frac{1}{F}\right). \tag{3.1}$$

When $1/s_o + 1/s_i = 1/F$, the path-dependent phase vanishes: contributions from all pupil positions arrive in phase at the image point.

3.2 Point Spread Function and Optical Transfer Function

Let $B(\boldsymbol{\xi})$ contain the aperture, amplitude weighting, and residual phase error, while the ideal lens phase has already been written explicitly. For an object point at $\boldsymbol{x}_o$ and an image-plane coordinate $\boldsymbol{x}$, the Fresnel calculation is

$$U_i(\boldsymbol{x};\boldsymbol{x}_o) \propto \int B(\boldsymbol{\xi})\exp\left\{\frac{\mathrm{i}k}{2}\left[\frac{|\boldsymbol{\xi}-\boldsymbol{x}_o|^2}{s_o} - \frac{|\boldsymbol{\xi}|^2}{F} + \frac{|\boldsymbol{x}-\boldsymbol{\xi}|^2}{s_i}\right]\right\}\mathrm{d}^2\xi. \tag{3.2}$$

Expand the squares. At the imaging condition, the $|\boldsymbol{\xi}|^2$ term cancels, leaving a pupil-independent phase $\Phi_0 = k(|\boldsymbol{x}_o|^2/s_o + |\boldsymbol{x}|^2/s_i)/2$ and

$$U_i \propto \mathrm{e}^{\mathrm{i}\Phi_0}\int B(\boldsymbol{\xi})\mathrm{e}^{-\mathrm{i}2\pi\boldsymbol{\xi}\cdot(\boldsymbol{x}-M\boldsymbol{x}_o)/(\lambda s_i)}\mathrm{d}^2\xi, \qquad M = -\frac{s_i}{s_o}. \tag{3.3}$$

Use reference-sphere field envelopes to remove these known ideal quadratic phases, and absorb constant gains and the coordinate Jacobian in the field convention. With the pupil-axis convention $\boldsymbol{f} = -\boldsymbol{\xi}/(\lambda s_i)$, the result is

$$\boxed{h(\boldsymbol{x}) = \int P(\boldsymbol{f})\mathrm{e}^{\mathrm{i}2\pi\boldsymbol{f}\cdot\boldsymbol{x}}\mathrm{d}^2f = \mathcal{F}^{-1}[P](\boldsymbol{x}), \qquad h_j(\boldsymbol{x}) = h(\boldsymbol{x}-M\boldsymbol{x}_{o,j}).} \tag{3.4}$$

h is the *amplitude* point-spread function (PSF): the image field from a unit point object. Its argument is a transverse image-plane displacement, not a propagation distance. The pupil integration variable $\boldsymbol{f}$ has inverse-length units; the condition is $|\boldsymbol{f}| \le f_c$, not $|\boldsymbol{x}| \le f_c$.

For a circular pupil of physical radius a, the paraxial mapping gives $f_c \simeq a/(\lambda s_i) = \mathrm{NA}/\lambda$. In the usual scalar pupil model,

$$\boxed{P(\boldsymbol{f}) = A(\boldsymbol{f})\,\mathbf{1}_{|\boldsymbol{f}|\le f_c}\,\exp\!\left[\frac{\mathrm{i}2\pi}{\lambda} W\!\left(\frac{\boldsymbol{f}}{f_c}\right)\right].} \tag{3.5}$$

$A \ge 0$ describes amplitude attenuation or apodization; W is phase-equivalent optical path error relative to ideal imaging, including any residual coating phase assigned to the projector. The disk describes the support only: an aberrated P is a *complex function on that disk*. Other pupil shapes give other supports.

Within a region where the system is approximately shift invariant, superposition gives, in image-equivalent object coordinates,

$$\boxed{U_i = h * U_o = \mathcal{F}^{-1}[P\widetilde{U}_o], \qquad I_{\mathrm{coh}} = |U_i|^2.} \tag{3.6}$$

If $U_{\mathrm{inc}} = 1$, then $U_o = t$ and $\widetilde{U}_o = \widetilde{t}$. $\widetilde{t}$ describes which field components the mask supplies; P describes how the projector changes them. For physical mask coordinates, first apply the magnification and orientation mapping. In a 4:1 reduction, $|M| = 1/4$; an image frequency is four times the corresponding physical mask frequency.

3.3 The Airy pattern and what an aberration changes

For a clear, uniformly weighted circular pupil, $P = 1$ within f_c. Writing $r = |\boldsymbol{x}|$ and using polar pupil coordinates,

$$\begin{aligned} h(r) &= \int_0^{f_c} f\,\mathrm{d}f \int_0^{2\pi} \mathrm{e}^{\mathrm{i}2\pi f r\cos\theta}\mathrm{d}\theta = 2\pi\int_0^{f_c} f J_0(2\pi f r)\mathrm{d}f \\ &= \frac{f_c J_1(2\pi f_c r)}{r}, \qquad h(0) = \pi f_c^2. \end{aligned} \tag{3.7}$$

The angular integral defines $2\pi J_0$; the radial integral uses $\mathrm{d}[vJ_1(v)]/\mathrm{d}v = vJ_0(v)$. Squaring the amplitude yields

$$\boxed{\frac{I(r)}{I(0)} = \left[\frac{2J_1(v)}{v}\right]^2, \qquad v = 2\pi f_c r, \qquad r_0 = \frac{3.8317}{2\pi f_c} = 0.6098\frac{\lambda}{\mathrm{NA}}.} \tag{3.8}$$

The first zero of J_1 defines the radius of the central Airy disk. At 13.5 nm and NA = 0.33, $r_0 = 24.95$ nm. Figure 2 shows why the rings look prominent on a logarithmic display but faint on the linear image: the first bright ring peaks at only 1.75% of the central intensity. All panels show the same physical PSF.

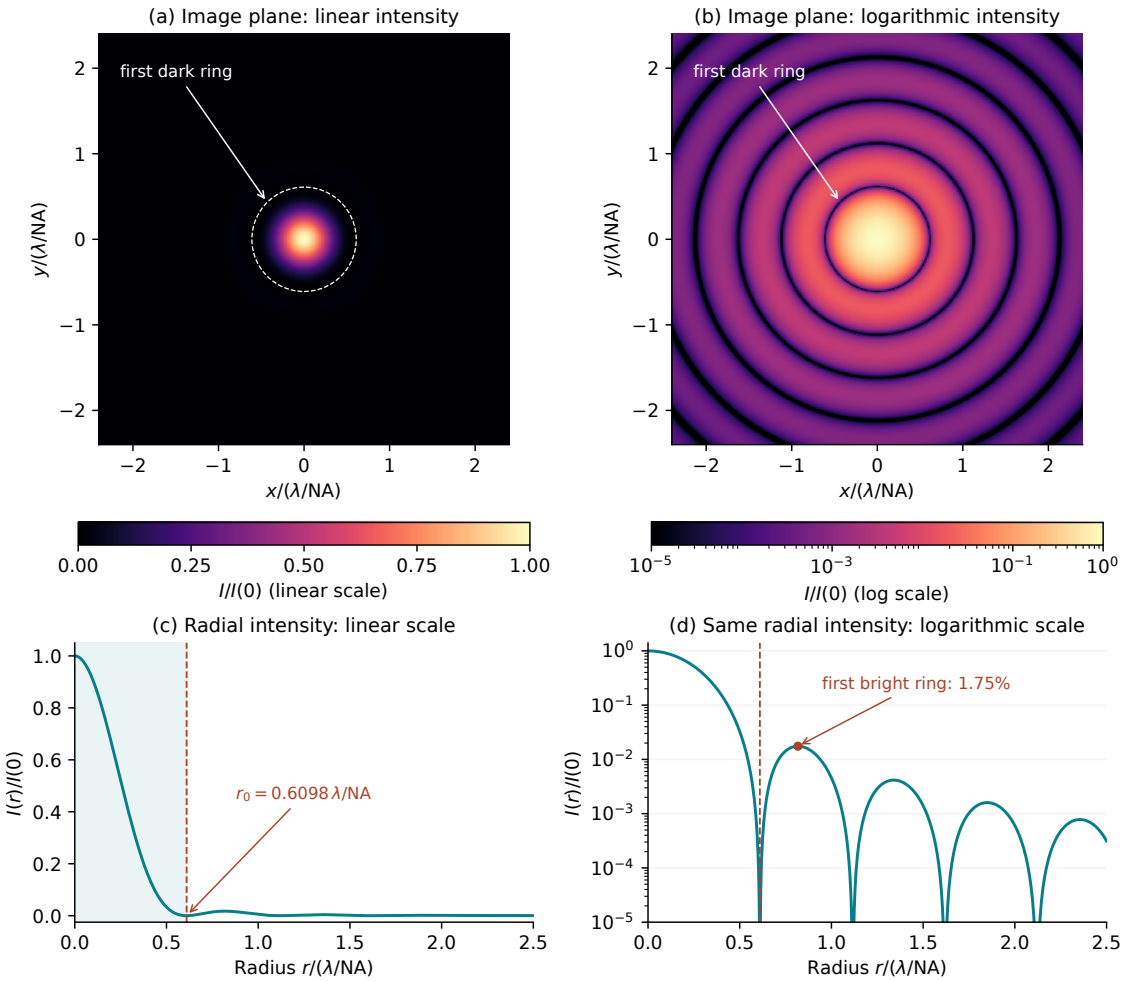


Figure 2: Airy intensity for a clear circular pupil. (a,b) The same image-plane intensity on linear and logarithmic scales; (c,d) radial profiles. The linear panels show unlogged $I/I(0)$; absolute irradiance requires a specified source strength. The dashed circle marks the first dark ring. Logarithmic displays stop at 10^{-5}; ideal minima are zero.

Aberration changes the phases in the pupil integral and therefore changes the PSF, even if the pupil intensity $|P|^2$ is unchanged. A constant phase (piston) changes no intensity; a linear pupil phase translates the PSF; quadratic phase produces defocus.

4 Coherence, interference averaging, and the doubled cutoff

4.1 Why coherent and incoherent images differ

Take two object-point amplitudes $a_1(\tau)$ and $a_2(\tau)$ whose image fields overlap at the *same* observation point: $U_i = a_1 h_1 + a_2 h_2$, where $h_j = h(\boldsymbol{x} - M\boldsymbol{x}_{o,j})$. After optical-cycle normalization, the exposure average gives

$$\begin{aligned} I_i &= \left\langle |a_1 h_1 + a_2 h_2|^2 \right\rangle \\ &= \left\langle |a_1|^2 \right\rangle |h_1|^2 + \left\langle |a_2|^2 \right\rangle |h_2|^2 + 2\,\mathrm{Re}\left[\underbrace{\langle a_1 a_2^* \rangle}_{\Gamma_{12}} h_1 h_2^* \right]. \end{aligned} \tag{4.1}$$

For fixed magnitudes, $\Gamma_{12} = A_1 A_2 \left\langle \mathrm{e}^{\mathrm{i}(\phi_1-\phi_2)} \right\rangle$. A stable relative phase preserves interference. For mutually incoherent contributions $\Gamma_{12} = 0$; for example, uniformly sampled relative phase gives $(2\pi)^{-1} \int_0^{2\pi} \mathrm{e}^{\mathrm{i}\phi} \mathrm{d}\phi = 0$. If each contribution alone has image intensity I_0, their sum has intensity $4I_0$ in phase, zero in antiphase, and $2I_0$ when mutually incoherent. *Averaging over the optical carrier alone does not remove interference between coherent waves of the same frequency.*

For many points,

$$I_i(\boldsymbol{x}) = \sum_{j,k} \langle a_j a_k^* \rangle \, h_j(\boldsymbol{x}) h_k^*(\boldsymbol{x}).$$

Mutually incoherent object points remove all $j \neq k$ terms, leaving a sum of intensity PSFs. Passing to a continuous object and calibrating out the overall throughput gives

$$\boxed{I_{\mathrm{incoh}} = p * I_o, \qquad p(\boldsymbol{x}) = \frac{|h(\boldsymbol{x})|^2}{\int |h(\boldsymbol{x}')|^2 \mathrm{d}^2 x'}, \qquad \int p \, \mathrm{d}^2 x = 1.} \tag{4.2}$$

This is linear in *object intensity*; Eq. (3.6) is linear in *object field*. The same point-response shape $|h|^2$ appears in both. Coherence changes how different point responses combine. For partial coherence the off-diagonal correlations are generally nonzero, so neither limiting convolution alone describes the image.

4.2 Deriving the unnormalized OTF as a pupil autocorrelation

Define the unnormalized intensity transfer function C and normalized optical transfer function T by

$$C(\boldsymbol{q}) = \mathcal{F}[|h|^2](\boldsymbol{q}), \qquad T(\boldsymbol{q}) = \frac{C(\boldsymbol{q})}{C(0)} = \mathcal{F}[p](\boldsymbol{q}). \tag{4.3}$$

Here $\boldsymbol{q}$ is an *intensity* spatial frequency. Insert $h = \int P(\boldsymbol{f}) \mathrm{e}^{\mathrm{i}2\pi \boldsymbol{f}\cdot\boldsymbol{x}} \mathrm{d}^2 f$:

$$\begin{aligned} C(\boldsymbol{q}) &= \iint P(\boldsymbol{f}_1) P^*(\boldsymbol{f}_2) \underbrace{\int \mathrm{e}^{\mathrm{i}2\pi(\boldsymbol{f}_1 - \boldsymbol{f}_2 - \boldsymbol{q})\cdot\boldsymbol{x}} \mathrm{d}^2 x}_{\delta^{(2)}(\boldsymbol{f}_1 - \boldsymbol{f}_2 - \boldsymbol{q})} \mathrm{d}^2 f_1 \mathrm{d}^2 f_2 \\ &= \int P(\boldsymbol{f}) P^*(\boldsymbol{f} - \boldsymbol{q}) \mathrm{d}^2 f \\ &= \boxed{\int P(\boldsymbol{f} + \boldsymbol{q}/2) P^*(\boldsymbol{f} - \boldsymbol{q}/2) \mathrm{d}^2 f.} \end{aligned} \tag{4.4}$$

At zero separation, Parseval's theorem yields

$$C(0) = \int |P(\boldsymbol{f})|^2 \mathrm{d}^2 f = \int |h(\boldsymbol{x})|^2 \mathrm{d}^2 x, \qquad T(0) = 1, \qquad \widetilde{I}_i(\boldsymbol{q}) = T(\boldsymbol{q}) \widetilde{I}_o(\boldsymbol{q}). \tag{4.5}$$

To express the autocorrelation as a convolution, define $P^\dagger(\boldsymbol{f}) = P^*(-\boldsymbol{f})$. Then

$$\boxed{C = P * P^\dagger.} \tag{4.6}$$

It is a convolution with a *conjugated and reflected* copy. For an unaberrated clear circular pupil, P is real and even, so the distinction disappears and we convolve two identical disk indicators.

4.3 Disk overlap explains the support and the contrast loss

The two factors in Eq. (4.4) are nonzero on disks of radius f_c with centres separated by $q = |\boldsymbol{q}|$. For the ideal unit-weight pupil, their product is one in the intersection and zero elsewhere. Consequently

$C(q)$ is the overlap area. For $0 \le q \le 2f_c$, let $\theta = \cos^{-1}(q/2f_c)$. The overlap is twice a circular sector minus its triangle:

$$C(q) = 2\left[f_c^2\theta - \frac{q}{2}\sqrt{f_c^2 - \frac{q^2}{4}}\right],$$
$$T(q) = \begin{cases} \dfrac{2}{\pi}\left[\cos^{-1}\nu - \nu\sqrt{1-\nu^2}\right], & 0 \le \nu \le 1, \\ 0, & \nu > 1, \end{cases} \qquad \nu = \frac{q}{2f_c}. \tag{4.7}$$

We divided by $C(0) = \pi f_c^2$. For $q > 2f_c$ there is no overlap; at $q = 2f_c$ the disks touch at one point of zero area. Thus the incoherent intensity cutoff is $2f_c = 2\mathrm{NA}/\lambda$.

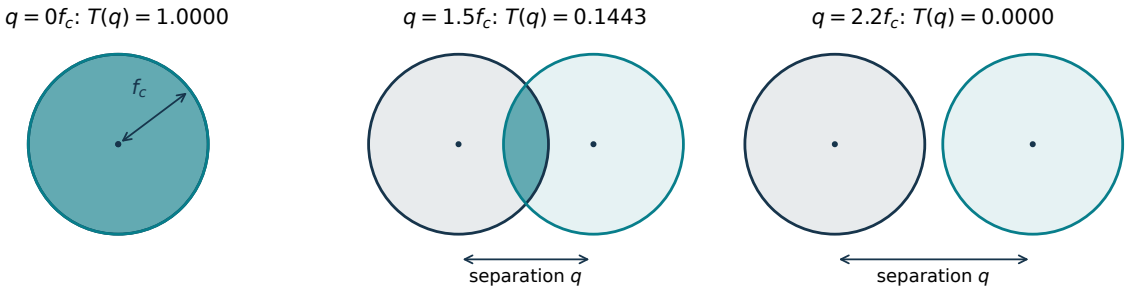


Figure 3: The unnormalized OTF of a clear circular pupil is the shaded overlap of two frequency disks, each of radius f_c. Their centre separation is the intensity frequency q. Dividing the overlap by πf_c^2 gives the values shown. There is no overlap beyond $2f_c$. These disks are in spatial-frequency space, not two physical holes in the optical system.

For an aberrated pupil, overlapping points also carry the phase factor

$$\exp\left\{\frac{\mathrm{i}2\pi}{\lambda}\left[W\left(\frac{\boldsymbol{f}+\boldsymbol{q}/2}{f_c}\right) - W\left(\frac{\boldsymbol{f}-\boldsymbol{q}/2}{f_c}\right)\right]\right\}.$$

The OTF is then a weighted complex overlap integral: phase cancellation can reduce transfer even when the supports overlap. Its modulus $|T|$ is the modulation transfer function (MTF); its phase describes phase shifts of intensity modulations. A cutoff alone does not tell us the contrast available below it.

The difference-frequency interpretation follows directly from Eq. (4.4): two admitted field components at $\boldsymbol{f}_1, \boldsymbol{f}_2$ produce an intensity term at $\boldsymbol{q} = \boldsymbol{f}_1 - \boldsymbol{f}_2$. Each field frequency lies within f_c, but their separation can approach $2f_c$.

4.4 A sinusoidal-mask example between the two cutoffs

Use an aberration-free circular objective in air, scalar paraxial imaging at best focus, $|M| = 1$, $\lambda = 500\,\mathrm{nm}$, and $\mathrm{NA} = 0.05$. Then $f_c = 0.100\,\mu\mathrm{m}^{-1}$. The thin mask, uniform in y, has

$$t(x) = t_0 + \epsilon\cos(2\pi q x), \qquad t_0 = 0.50, \quad \epsilon = 0.05, \quad q = 1.5f_c, \qquad p = 1/q = 6.67\,\mu\mathrm{m}. \tag{4.8}$$

This is *amplitude* transmission; its intensity transmission is $|t|^2$. The mask is periodic, with no finite-edge effects.

A: coherent, normal illumination. The field orders are $0, +q, -q$, with amplitudes $t_0, \epsilon/2, \epsilon/2$. Both sidebands lie outside the pupil:

$$U_A = t_0 P(0) + \frac{\epsilon}{2}P(q)\mathrm{e}^{\mathrm{i}2\pi q x} + \frac{\epsilon}{2}P(-q)\mathrm{e}^{-\mathrm{i}2\pi q x} = t_0, \qquad \boxed{I_A = 0.25.} \tag{4.9}$$

B: an illumination giving the incoherent imaging response. Choose mutually incoherent plane-wave directions uniformly filling $|\boldsymbol{s}| \le 3f_c$ (illumination NA 0.15). For this three-order mask, every source direction admitting any order lies within $|\boldsymbol{s}| \le q + f_c = 2.5f_c$; the uniform source therefore covers every relevant overlap region. After calibration, its source average gives the full incoherent OTF for this example. This finite source need not be spatially incoherent at every separation for every possible object.

Calibrate both cases to the response $t_0^2 = 0.25$ of a uniform mask $t = t_0$. At equal incident power, B's uniform-mask throughput is 1/9 of A's in this model, so B is divided by that factor. The comparison concerns contrast after calibration, not photon efficiency.

Expand the object intensity and apply T to its Fourier components:

$$\begin{aligned} |t(x)|^2 &= t_0^2 + \frac{\epsilon^2}{2} + 2t_0\epsilon\cos(2\pi qx) + \frac{\epsilon^2}{2}\cos(4\pi qx), \\ T(q) &= \frac{2}{\pi}\left[\cos^{-1}(0.75) - 0.75\sqrt{1-0.75^2}\right] = 0.144294, \qquad T(2q) = 0, \\ I_B(x) &= 0.25125 + 0.00721468\cos(2\pi qx). \end{aligned} \tag{4.10}$$

The intensity modulation at $q < 2f_c$ survives; the harmonic at $2q = 3f_c$ is removed. The $\epsilon^2/2$ term explains B's slightly higher mean. Using Michelson contrast $\mathcal{V} = (I_{\max} - I_{\min})/(I_{\max} + I_{\min})$, the mask has 19.80% contrast, A has 0, and B has 2.87% (Fig. 4). This is a concrete case of higher cutoff preserving finer periodic structure, though with strongly attenuated contrast.

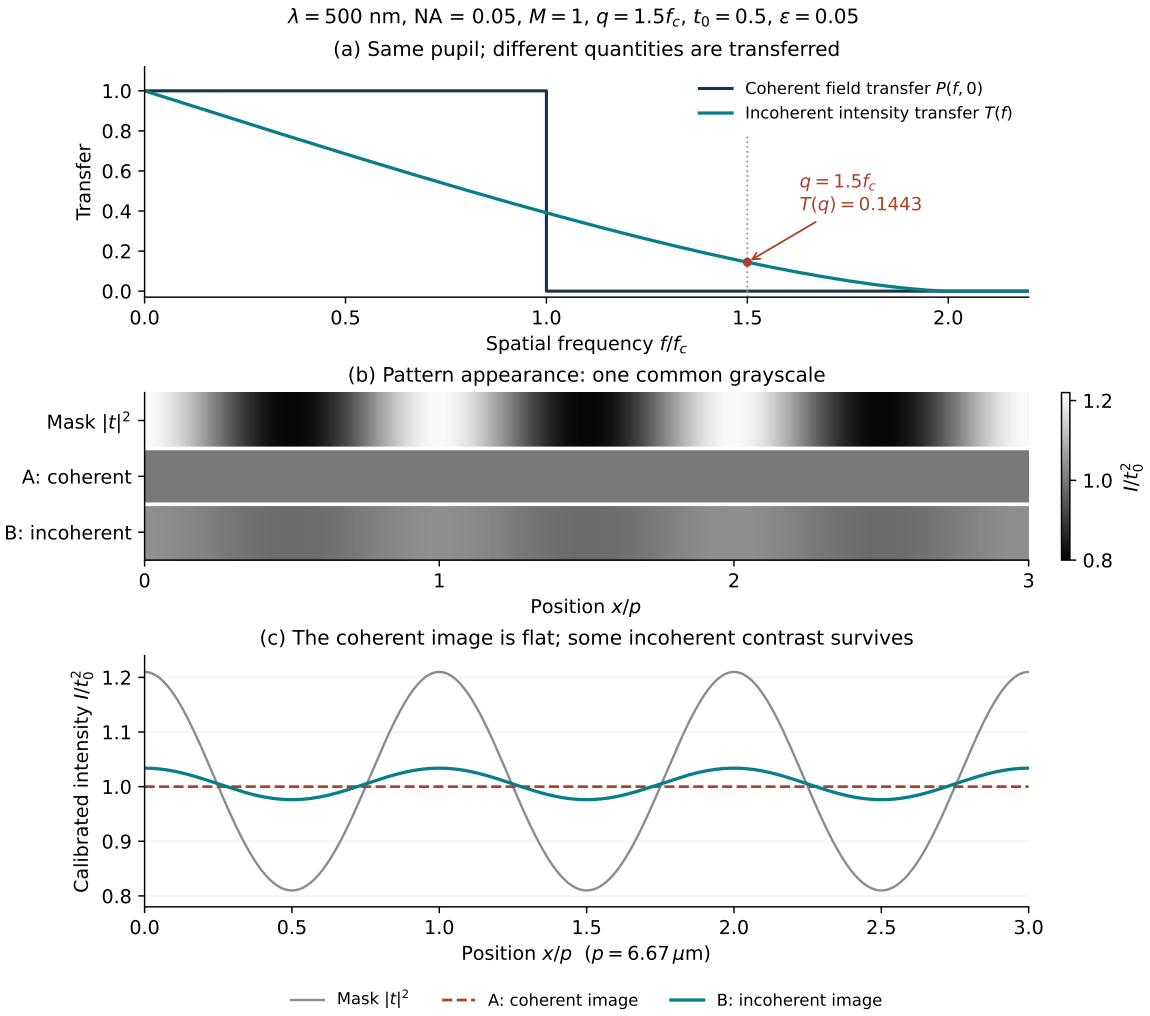


Figure 4: Sinusoidal-mask comparison with the same objective. (a) Coherent *field* transfer and incoherent *intensity* transfer. (b) One common grayscale, without separate contrast stretching. (c) Profiles divided by t_0^2 after uniform-mask calibration. A uses normal coherent illumination; B uses mutually incoherent source directions uniformly filling a disk of radius $3f_c$.

To see where B's modulation comes from, a source direction $s_x = -q/2$ places the zero and $+q$ orders at $-0.75f_c$ and $+0.75f_c$. Both fit inside the pupil and interfere at their difference frequency q. The source average retains some modulation. Appropriately oblique *coherent* illumination can also admit this pair; the result above compares B specifically with *normal* coherent illumination.

References

[1] M. Born and E. Wolf, *Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light*, 4th ed., Pergamon Press. See §1.3 for electromagnetic waves, §§1.5–1.6 for reflection and films, §§8.3 and 8.5.2 for diffraction and the circular aperture, and §9.5 and §10.4 for imaging and coherence.

[2] B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, 2nd ed., Wiley (2007). See §§2.1–2.2 and 2.5 for wave and paraxial optics, §§4.1–4.4 for Fourier optics and image formation, and §6.2 for electromagnetic reflection.

[3] Z. Chen, L. Dong, H. Ding, and Y. Wei, "Aberration budget analysis of EUV lithography from the imaging performance of a contact layer in a 5 nm technology node," *Applied Optics* **62**(27), 7270–7279 (2023). doi:10.1364/AO.496461.